Causal Inference

Session 2 — Randomised experiments

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Albert School · Master 2

Academic year 2026–2027

ALBERT SCHOOL

Where we left Session 1

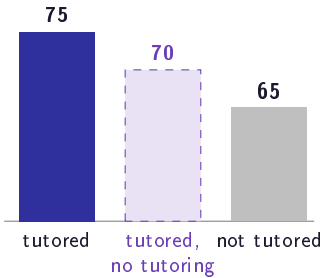
Introduction

OVERVIEW

The raw gap mixed a real effect with a head start.

Tutored volunteers scored 75, the others 65. But the volunteers would have scored 70 anyway: 5 points of effect, 5 points of head start.

Today's question: *can we design the study so that the head start disappears?*



Remember. Session 1 diagnosed the problem. Session 2 fixes it by design.

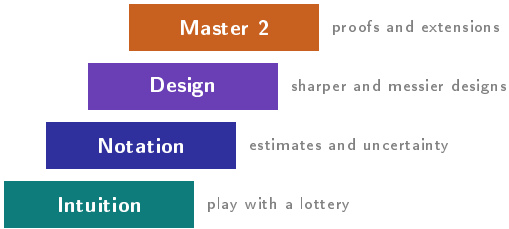
How this session is built

Introduction

OVERVIEW

Same four steps as last week: stories and numbers first, proofs last.

We first *play* with a lottery on the six students of Session 1, and see with our own eyes what randomisation does. Then we name what we saw, plan an experiment, improve its design, and handle what goes wrong. The Master 2 part proves every claim made along the way.



Remember. Every formula today describes a lottery you will already have replayed by hand.

By the end of today you should be able to

Introduction OVERVIEW

1. Explain why a lottery removes selection bias, and what chance still leaves behind.
2. Compute a difference in means, its standard error and a confidence interval.
3. Plan a sample size from a smallest meaningful effect and a target power.
4. Improve precision with blocking, CUPED and regression adjustment; handle clusters.
5. Recognise noncompliance, attrition, peeking and multiple testing, and respond to each.
6. Derive Neyman's variance, CUPED optimality, design effects and exact randomisation tests.

Remember. Items 1–3 intuition and notation, 4–5 design, 6 Master 2 mathematics.

Session 2 roadmap

1. Why a lottery?	**Intuition**
2. Replaying the lottery	Intuition
3. From the lottery to estimates	Notation
4. Planning: power and sample size	Notation
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6. When the experiment is not clean	Design
7. The Master 2 toolkit	Master 2
8. Practice and wrap-up	Synthesis

Too few places, so the school draws lots

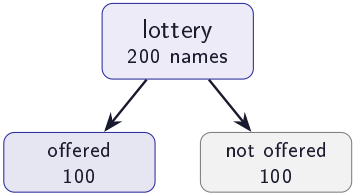
Why a lottery?

INTUITION

The school has 100 tutoring places for 200 eligible students.

Instead of letting volunteers sign up, it draws names from a hat. At the end of term it compares exam results of the two groups.

Nobody chose their group. Not the motivated students, not the teachers, not the parents.



Remember. A lottery takes the choice of treatment away from the people being studied.

Why this kills the head start

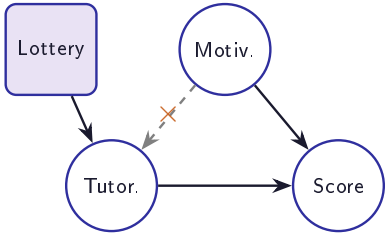
Why a lottery?

INTUITION

In Session 1, motivation pushed students into tutoring.

With a lottery, motivation can still raise scores, but it no longer decides who is tutored. The arrow from motivation to tutoring is cut.

Motivated students end up in both groups in roughly equal shares, so their head start is shared too.



Remember. This is the "doing" graph of Session 1, produced in real life.

What the lottery does not change

Why a lottery? INTUITION

Students keep their motivation, ability and family background. Only the offer is drawn by lot.

The lottery does not look at anyone's potential scores, so the two groups are alike *on average*: same share of motivated students, same share of strong students, same everything.

But "on average" means over many possible draws. One particular draw with few students can still be unlucky, for example putting more strong students in one group by chance.

Remember. Randomisation removes systematic selection, not every chance imbalance.

Random assignment is not random sampling

Why a lottery? INTUITION

Two different uses of chance solve two different problems.

	Random **assignment**	Random **sampling**
What is drawn?	who gets treated	who is in the study
What it buys	a fair comparison	a representative study
Validity	internal	external

An experiment on volunteers from one school can be perfectly fair internally and still say little about other schools.

Remember. A fair comparison need not represent everyone.

Which unit should be drawn?

Why a lottery? INTUITION

If tutored students share notes with classmates, individual lotteries leak.

Drawing **students** gives many independent draws, but untutored classmates may benefit indirectly, shrinking the measured gap.

Drawing **classes** keeps treated and untreated students apart, but leaves fewer independent draws, hence more noise (Part 5).

The right choice follows how the intervention spreads.

Remember. Randomise at the level where the treatment actually operates.

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Back to our six students

Replaying the lottery

INTUITION

We reuse the oracle table of Session 1: both worlds are known, for teaching only.

Student	Score if tutored	Score if not tutored	Effect
Awa	75	70	5
Léo	76	72	4
Inès	74	68	6
Moussa	71	62	9
Chloé	72	66	6
Karim	70	67	3

The true average effect is 5.5. Now three places are drawn by lot.

Remember. Knowing the truth lets us watch what a lottery does to our estimate.

One draw of the lottery

Replaying the lottery

INTUITION

Suppose the hat picks Awa, Chloé and Karim for tutoring.

We see their tutored scores and the others' untutored scores:

$$\underbrace{\frac{75+72+70}{3}}_{\text{tutored: } 72.3} - \underbrace{\frac{72+68+62}{3}}_{\text{not tutored: } 67.3} = 5.0.$$

Close to 5.5, but not equal. A different draw would give a different number. How different?

Remember. Each draw gives one estimate; the draw itself is random.

All 20 possible lotteries

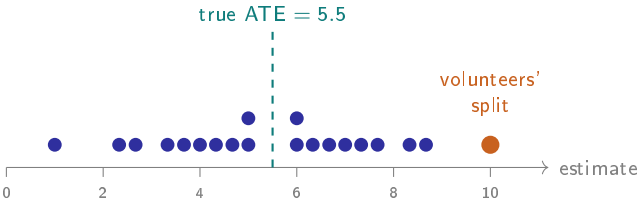
Replaying the lottery

INTUITION

There are 20 ways to choose 3 students out of 6. Each dot is one of them.

The estimates range from 1 to 10. Their average is exactly **5.5**, the true effect.

The orange dot is the volunteers' split of Session 1: it sits at the very edge of the cloud, tied with the mirror split for the largest error.



Remember. A lottery is right on average (unbiased) but noisy in any single draw.

Two ideas from one picture

Replaying the lottery

INTUITION

Everything else in this session grows from these two observations.

1. Centre: no bias.
The cloud of estimates is centred on the truth. Volunteering pushed the estimate to one side; the lottery does not.

2. Spread: noise.
The cloud is wide. With only six students, a single draw can be far from 5.5. We need to measure and shrink that spread.

Part 3 names the centre (unbiasedness) and the spread (standard error). Parts 4–5 shrink the spread.

Remember. Randomisation handles the centre; design and sample size handle the spread.

More students, narrower cloud

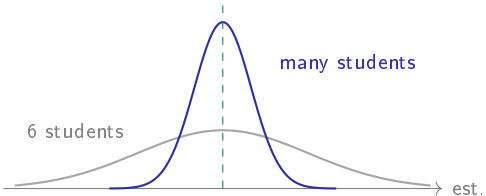
Replaying the lottery

INTUITION

With 200 students, lucky and unlucky draws average out inside each group.

The cloud of possible estimates stays centred on the truth but becomes much narrower. A single experiment is then very likely to land close to the true effect.

That is why a real trial uses many students, not six.



Remember. Sample size does not change the centre; it narrows the spread.

Checkpoint

Replaying the lottery

INTUITION

Discuss with your neighbour for three minutes.

1. Why does the volunteers' split sit at the edge of the dot plot?
2. A colleague says: "our lottery gave 8, so the true effect is 8." What is wrong?
3. A second colleague says: "lotteries are useless with small samples." Is that right?

Remember. Unbiased does not mean correct every time; it means correct on average over draws.

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A real readout

From the lottery to estimates

NOTATION

A constructed trial with full compliance and complete follow-up.

Assigned group	Students	Mean score	Standard deviation
Tutoring	100	74	10
Control	100	69	10

The gap is 5 points. Thanks to the lottery, it has no head start built in. Our remaining job: how far could this 5 be from the truth?

Remember. An experiment gives an unbiased estimate, not an exact answer.

The difference in means

From the lottery to estimates NOTATION

What we did by hand with Awa, Chloé and Karim, written with symbols.

$$\hat{\tau} = \bar{Y}_1 - \bar{Y}_0$$

$\bar{Y}_1$: average observed score in the tutored group; $\bar{Y}_0$: in the control group. The hat means "estimated from data". Here $\hat{\tau} = 74 - 69 = 5$.

In Session 1 this same formula was biased. Nothing in the formula changed: the *design* changed.

Remember. A simple estimator works because the design does the causal work.

Unbiasedness, in symbols

From the lottery to estimates

NOTATION

“The dot plot is centred on the truth” becomes:

$$E_D\left[\hat{\tau}\right] = \frac{1}{N}\sum_{i=1}^{N}\left\{Y_i(1) - Y_i(0)\right\} = \tau$$

E_D averages over all possible lottery draws D, keeping the students and their potential scores fixed. For our six students, the average of the 20 dots is exactly 5.5.

Why? Each student is tutored in the same share of draws, so each group mean is right on average.

Remember. Randomness comes from the lottery, not from the students.

The short algebra behind unbiasedness

From the lottery to estimates

NOTATION

With N_1 places among N students, each student is drawn with probability N_1/N.

$$E_D\left[\frac{1}{N_1}\sum_i D_i Y_i(1)\right] = \frac{1}{N_1}\sum_i \frac{N_1}{N} Y_i(1) = \frac{1}{N}\sum_i Y_i(1).$$

The expected tutored-group mean equals everyone's average score *if tutored*. The same argument gives the control group, and subtracting yields τ.

Compare with Session 1: the selection-bias term $E[Y(0) \mid D = 1] - E[Y(0) \mid D = 0]$ is now zero by construction.

Remember. The lottery makes the treated group a fair sample of everyone's treated world.

The standard error measures the width of the cloud

From the lottery to estimates NOTATION

The standard error is the typical distance between one draw's estimate and the truth.

It describes how much $\hat{\tau}$ would vary if we re-ran the lottery. It is *not* the spread of students' exam scores.

Two sources make it large: scores that vary a lot between students, and few students per group.

Remember. Outcome variability and estimator uncertainty are different quantities.

Computing the standard error

From the lottery to estimates

NOTATION

Standard deviation 10 in each arm, 100 students per arm.

$$\widehat{SE} = \sqrt{\frac{s_1^2}{n_1} + \frac{s_0^2}{n_0}} = \sqrt{\frac{100}{100} + \frac{100}{100}} \approx 1.41$$

s_d: standard deviation of scores within arm d; n_d: its size. Each group mean has its own noise, and the noises add up in the difference.

Remember. More students per arm make each group mean more stable.

A confidence interval

From the lottery to estimates

NOTATION

Estimate 5, standard error 1.41.

An approximate 95% interval is

$$5 \pm 1.96 \times 1.41 = [2.23,\ 7.77].$$

The *procedure* covers the true effect in about 95% of repeated lotteries. The realised interval is the range of effects compatible with this experiment. It excludes zero, and also excludes effects above 8.

Remember. Report both the estimate and how precisely it was learned.

Significant is not the same as important

From the lottery to estimates NOTATION

An effect can be precisely estimated and still be too small to justify the cost.

Before looking at the data, fix the smallest improvement that would matter to the school, say 3 points. Then read the whole interval, weigh the cost of tutoring and possible harms.

A small p-value only says that zero is implausible. It does not say the programme is worth its budget.

Remember. The decision threshold should reflect the school's objective, not a 5% convention.

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Would the experiment notice a real effect?

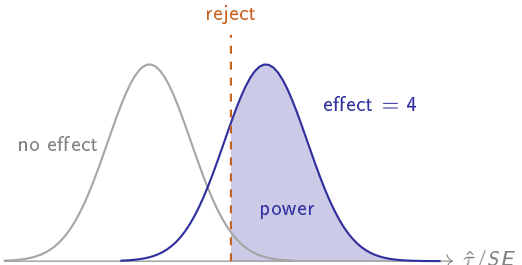
Planning: power and sample size

NOTATION

Before running the trial, imagine tutoring really adds 4 points.

Grey: the cloud of estimates if the effect were zero. The dashed line is where we would call a result "significant".

Blue: the cloud if the effect is 4. The shaded share of it crosses the line: that share is the **power**.



Remember. Power is the chance of detecting an effect that really exists; aim for about 80%.

Why small effects need large samples

Planning: power and sample size

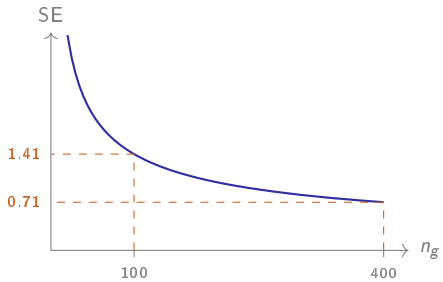
NOTATION

Noise shrinks slowly as we add students.

With n_g students per arm and score standard deviation σ,

$$SE \approx \sigma\sqrt{2/n_g}.$$

Halving the standard error needs *four* times as many students. So detecting an effect half as large needs about four times the sample.



Remember. Precision grows with the square root of the sample size.

The planning formula

Planning: power and sample size NOTATION

Decide three things first: the false-alarm rate, the power, and the smallest effect worth detecting.

$$n_g \approx \frac{2\sigma^2 \left(z_{1-\alpha/2} + z_{1-\beta}\right)^2}{\delta^2}$$

δ: effect to detect; α: false-positive rate (5%, $z = 1.96$); $1 - \beta$: power (80%, $z = 0.84$).
Example: $\sigma = 10$, $\delta = 4$ gives $n_g \approx 2 \cdot 100 \cdot 2.8^2/16 \approx 98$ students per arm.

Remember. Sample-size inputs are substantive choices, not software defaults.

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Blocking: avoid unlucky draws in advance

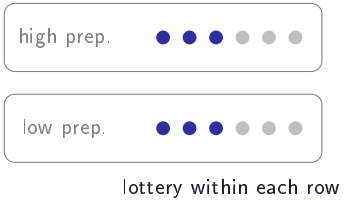
Sharper designs

DESIGN

If preparation strongly predicts scores, why leave its balance to chance?

First split students into preparation groups (blocks). Then run a separate lottery inside each block. Both arms now contain the same share of strong and weak students by design.

Estimate the effect in each block and average with the block shares, exactly as in Session 1's standardisation.



Remember. Blocking balances key baseline variables before outcomes occur.

Using last year's scores to remove noise

Sharper designs DESIGN

Part of every student's final score was predictable before tutoring even started.

A strong student scores high in both years. That predictable part adds noise to the comparison without saying anything about tutoring.

Idea: subtract the part of the final score that last year's score predicts, then compare the groups. Use only information measured *before* the lottery, never study habits changed by tutoring.

Remember. A fair comparison can also be made more precise.

CUPED: compare what could not be predicted

Sharper designs

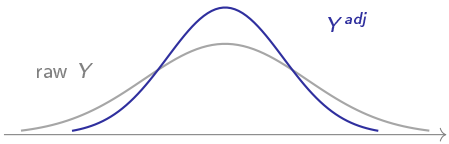
DESIGN

Adjusted score = final score − its predictable baseline part:

$$Y^{adj} = Y - \theta\,(X - \bar{X}),$$

where X is last year's score.

The adjusted scores are less spread out, so the two group means are less noisy. The lottery still guarantees the comparison is fair.



Remember. CUPED uses earlier information to shrink noise, not to change the target.

How much to subtract, and what we gain

Sharper designs DESIGN

Too little leaves predictable noise; too much adds new noise.

The best coefficient is the slope of Y on X:

$$\theta^* = \frac{\mathrm{Cov}(Y, X)}{\mathrm{Var}(X)}, \qquad \frac{\mathrm{Var}(Y^{adj})}{\mathrm{Var}(Y)} = 1 - \rho^2.$$

With correlation $\rho = 0.7$: variance ratio $1 - 0.49 = 0.51$, standard error multiplied by $\sqrt{0.51} \approx 0.71$. Our SE of 1.41 would drop to about 1.01.

Remember. A 49% variance reduction is a 29% standard-error reduction, for free.

Regression adjustment in practice

Sharper designs DESIGN

Students with different baseline scores may respond differently to tutoring.

Regress the outcome on treatment, the *centred* baseline score, and their interaction:

$$Y_i = \alpha + \tau D_i + \beta'(X_i - \bar{X}) + \gamma' D_i(X_i - \bar{X}) + u_i.$$

Centring makes τ the average effect; the interaction lets each arm have its own slope (Lin, 2013). Use heteroskedasticity-robust standard errors.

Remember. Adjustment should respect possible differences between arms.

Classes behave like clusters

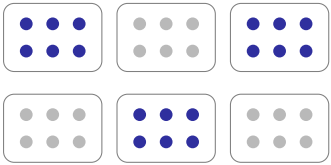
Sharper designs

DESIGN

Twenty students with one teacher share shocks that twenty strangers do not.

A good teacher, a noisy classroom, a flu wave: everybody in the class moves together. When whole classes are drawn, their students are not independent pieces of evidence.

Counting them as independent overstates what we know.



6 classes = 6 draws, not 36

Remember. Many correlated rows can contain few independent comparisons.

The design effect in numbers

Sharper designs

DESIGN

Class size $m = 25$, within-class correlation $\rho = 0.08$.

$$\text{design effect} = 1 + (m - 1)\rho = 1 + 24 \times 0.08 = 2.92$$

The variance is almost three times what an independent-student calculation claims. Equivalently, 1,000 clustered students carry the information of about 340 independent ones.

Adding classes helps far more than adding students inside the same classes.

Remember. Plan power, and compute standard errors, at the level of randomisation.

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Some students decline their place

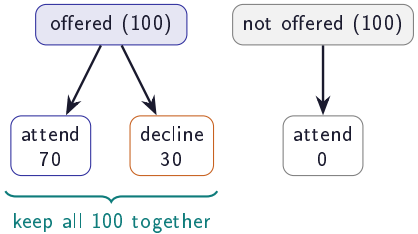
When the experiment is not clean

DESIGN

The lottery decides who is *offered* tutoring. Attending is still voluntary.

Comparing attendees with non-attendees brings the volunteers' head start straight back. Instead, compare by **original assignment**: the intention-to-treat effect (ITT).

ITT is the effect of the offer. Session 5 will rescale it into an effect of attending.



Remember. Analyse as randomised: the offer is what the lottery made fair.

Missing exam scores can undo the lottery

When the experiment is not clean DESIGN

Suppose struggling control students skip the final exam more often.

Then the control students we observe are the stronger ones, and the gap shrinks for reasons that have nothing to do with tutoring. The lottery made the groups comparable at the start; attrition can break that comparability at the end.

Always report follow-up rates by arm, and state the assumption used to handle missing scores.

Remember. Randomisation does not protect against selective loss of outcomes.

Peeking at the results every day

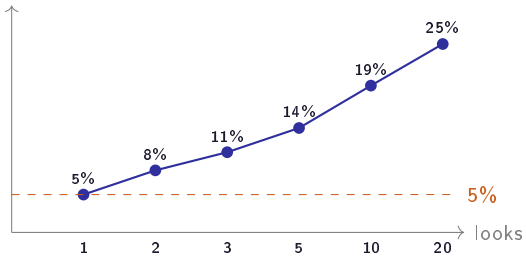
When the experiment is not clean

DESIGN

A team tests every day and stops at the first "significant" result.

Even with no effect at all, noise sometimes wanders past the 5% line. Each extra look is another chance for a false alarm.

With 20 looks, the real false-positive rate is about 25%, not 5%.



Remember. The stopping rule is part of the experiment. Simulated, equally spaced looks.

Monitoring safely

When the experiment is not clean

DESIGN

Stopping early for clear benefit or harm is legitimate, if the rule is planned.

Three valid options: a single analysis date fixed in advance; group-sequential boundaries that spend the 5% across planned looks; or confidence sequences that remain valid at any stopping time.

Choose the rule *before* seeing any treatment difference, and write it down.

Remember. Early stopping is fine when the inference accounts for the monitoring.

Twenty outcomes, twenty chances to be lucky

When the experiment is not clean

DESIGN

Test 20 unrelated outcomes at 5% each, and a false discovery is likely.

If tutoring does nothing and tests are independent, the chance of at least one "significant" result is

$$1 - 0.95^{20} \approx 64\%.$$

Remedy: name one **primary** outcome in advance. For m confirmatory tests, Bonferroni tests each at α/m. Label everything else as exploratory.

Remember. More opportunities to claim success require a multiplicity plan.

Fisher's idea: replay the lottery on the data

When the experiment is not clean

DESIGN

Suppose tutoring changed nobody's score. Then every student would have the same score whatever the draw.

Our draw: Awa, Chloé, Karim tutored, observed gap 5.0. Under "no effect for anyone", re-run all 20 lotteries on the *observed* scores and recompute the gap each time.

6 of the 20 re-draws give a gap at least as large as 5 in absolute value: $p = 6/20 = 0.30$. With six students, a gap of 5 is not surprising under no effect.

Remember. A known lottery generates its own reference distribution, with no normal approximation.

What belongs in an experiment report

When the experiment is not clean

DESIGN

A decision-maker needs the result and how it was produced.

Eligibility and recruitment; the unit and mechanism of assignment; the primary outcome and pre-specified analysis; the estimate with its interval; compliance and attrition by arm; the monitoring rule; possible spillovers; and a clear separation of confirmatory from exploratory results.

Remember. A credible readout connects the estimate to the actual experiment.

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The design-based viewpoint

The Master 2 toolkit MASTER 2

From intuition: "keep the students fixed and replay the lottery" — the 20 dots.

Potential outcomes $\{Y_i(1), Y_i(0)\}_{i=1}^{N}$ are fixed constants. The only randomness is the assignment vector $D = (D_1, \ldots, D_N)$. Under **complete randomisation**, D is uniform on vectors with exactly N_1 ones.

Define $\bar{Y}(d) = N^{-1} \sum_i Y_i(d)$, $S_d^2 = (N-1)^{-1} \sum_i (Y_i(d) - \bar{Y}(d))^2$, $\tau_i = Y_i(1) - Y_i(0)$, and S_τ^2 the finite-population variance of the τ_i.

Remember. Inference is about the lottery, not about a hypothetical super-population.

Assignment moments

The Master 2 toolkit MASTER 2

From intuition: drawing 3 names out of 6 means one student's place is another's missed chance.

Let $p = N_1/N$. Uniform assignment of exactly N_1 units gives

$$E[D_i] = p, \qquad E[D_i D_j] = \frac{N_1(N_1-1)}{N(N-1)} \quad (i \neq j).$$

Therefore

$$\mathrm{Var}(D_i) = p(1-p), \qquad \mathrm{Cov}(D_i, D_j) = -\frac{p(1-p)}{N-1}.$$

Remember. Assignment indicators are negatively dependent under a fixed number of places.

Variance of one arm's mean

The Master 2 toolkit MASTER 2

From intuition: each arm is a sample drawn without replacement from the class.

Put $a_i = Y_i(1) - \bar{Y}(1)$, so $\sum_i a_i = 0$ and $\sum_{i \neq j} a_i a_j = -\sum_i a_i^2$. Using the assignment moments,

$$\begin{aligned} \mathrm{Var}_D(\bar{Y}_1^{obs}) &= \frac{p(1-p)}{N_1^2}\Big[\sum_i a_i^2 - \frac{\sum_{i \neq j} a_i a_j}{N-1}\Big] \\ &= \frac{p(1-p)N}{N_1^2} S_1^2 = \Big(\frac{1}{N_1} - \frac{1}{N}\Big) S_1^2. \end{aligned}$$

Remember. Sampling without replacement yields the finite-population correction.

Covariance between the two arm means

The Master 2 toolkit MASTER 2

From intuition: the tutored and control groups are complements: fill one, you empty the other.

Let $S_{10} = (N-1)^{-1} \sum_i (Y_i(1) - \bar{Y}(1))(Y_i(0) - \bar{Y}(0))$. Expanding with $\mathrm{Cov}(D_i, 1 - D_j)$,

$$\mathrm{Cov}_D(\bar{Y}_1^{obs}, \bar{Y}_0^{obs}) = -\frac{S_{10}}{N}.$$

The variance of individual effects satisfies

$$S_\tau^2 = S_1^2 + S_0^2 - 2S_{10}.$$

Remember. The unobserved pairing of potential outcomes enters through their covariance.

Neyman's variance formula

The Master 2 toolkit MASTER 2

From intuition: the exact width of the cloud of 20 dots.

Combine the two variances and the covariance:

$$\begin{aligned}\mathrm{Var}_D(\hat{\tau}) &= \Big(\frac{1}{N_1} - \frac{1}{N}\Big) S_1^2 + \Big(\frac{1}{N_0} - \frac{1}{N}\Big) S_0^2 + \frac{2S_{10}}{N} \\ &= \frac{S_1^2}{N_1} + \frac{S_0^2}{N_0} - \frac{S_\tau^2}{N}.\end{aligned}$$

Every term follows from the known assignment distribution. No outcome model is used.

Remember. Effect heterogeneity enters exact uncertainty through a cross-world quantity.

Check on the six students

The Master 2 toolkit MASTER 2

From intuition: we can verify the formula because the oracle table reveals both worlds.

From the table: $S_1^2 = 5.6$, $S_0^2 = 11.9$, $S_\tau^2 = 4.3$, $N_1 = N_0 = 3$, $N = 6$.

$$\mathrm{Var}_D(\hat{\tau}) = \frac{5.6}{3} + \frac{11.9}{3} - \frac{4.3}{6} \approx 5.117.$$

The empirical variance of the 20 lottery estimates is exactly 5.117 (standard deviation ≈ 2.26, the width of the grey curve earlier).

In real data S_τ^2 is unknown: it requires both outcomes for each student.

Remember. Theory and brute-force enumeration agree to the last digit.

Conservative variance estimation

The Master 2 toolkit MASTER 2

From intuition: the usual $s_1^2/n_1 + s_0^2/n_0$ from Part 3.

Within-arm sample variances are unbiased for S_d^2, so with $\hat{V} = s_1^2/N_1 + s_0^2/N_0$,

$$E_D[\hat{V}] - \mathrm{Var}_D(\hat{\tau}) = \frac{S_\tau^2}{N} \geq 0.$$

$\hat{V}$ is exact when effects are constant ($S_\tau^2 = 0$), conservative otherwise. Combined with variance consistency and a CLT, Wald intervals are asymptotically conservative; the inequality alone does not prove finite-sample coverage.

Remember. Conservative coverage needs a distributional argument, not just a variance inequality.

A finite-population central limit theorem

The Master 2 toolkit MASTER 2

From intuition: "with many students the cloud looks like a bell curve".

Consider a sequence with $N \to \infty$, $N_1/N \to p \in (0,1)$, stable positive limiting variances and nondegenerate $N\,\mathrm{Var}_D(\hat{\tau})$. A sufficient no-dominant-unit condition is

$$\max_i \frac{|Y_i(d) - \bar{Y}(d)|^2}{N} \to 0 \qquad (d = 0, 1).$$

Then $(\hat{\tau} - \tau_N)/\sqrt{\mathrm{Var}_D(\hat{\tau})} \Rightarrow N(0,1)$ (Hájek; Li and Ding, 2017).

Remember. A large sample is not enough if a few units dominate the variation.

Exact Gaussian power

The Master 2 toolkit MASTER 2

From intuition: the shaded share of the blue curve in Part 4.

If $\hat{\tau}/SE \sim N(\kappa, 1)$ with $\kappa = \delta/SE$ and $c = z_{1-\alpha/2}$, both rejection tails give

$$\text{Power}(\delta) = 1 - \Phi(c - \kappa) + \Phi(-c - \kappa).$$

Ignoring the negligible opposite tail and setting $1 - \Phi(c - \kappa) = 1 - \beta$ gives $\kappa = c + z_{1-\beta}$; substituting $SE = \sigma\sqrt{2/n_g}$ recovers the planning formula.

Remember. The sample-size formula approximates a specified rejection probability.

Optimal allocation between arms

The Master 2 toolkit MASTER 2

From intuition: "equal groups" is a choice, and sometimes not the best one.

For a fixed total n, minimise $\sigma_1^2/n_1 + \sigma_0^2/(n-n_1)$. The first-order condition gives

$$\frac{\sigma_1^2}{n_1^2} = \frac{\sigma_0^2}{n_0^2} \quad \Longrightarrow \quad \frac{n_1}{n_0} = \frac{\sigma_1}{\sigma_0}$$

(Neyman allocation). The objective is convex in n_1, so this is the minimum. With per-unit costs c_d under a budget, $n_1/n_0 = (\sigma_1/\sigma_0)\sqrt{c_0/c_1}$.

Remember. Allocation can raise power without adding students.

Stratified estimation

The Master 2 toolkit MASTER 2

From intuition: a separate lottery in each preparation block.

Stratum h has N_h units, weight $w_h = N_h/N$, estimate $\hat{\tau}_h$. With independent lotteries across strata,

$$\hat{\tau}_{str} = \sum_h w_h \hat{\tau}_h, \qquad E_D[\hat{\tau}_{str}] = \sum_h w_h \tau_h = \tau,$$

$$\mathrm{Var}_D(\hat{\tau}_{str}) = \sum_h w_h^2 \Big(\frac{S_{1h}^2}{N_{1h}} + \frac{S_{0h}^2}{N_{0h}} - \frac{S_{\tau h}^2}{N_h} \Big).$$

Only *within*-stratum variances appear: variation explained by the strata is removed.

Remember. Stratification keeps the target and removes between-block noise.

CUPED optimality

The Master 2 toolkit MASTER 2

From intuition: "too little leaves noise, too much adds noise" is a parabola.

Write $c = \mathrm{Cov}(Y, X)$, $v_X = \mathrm{Var}(X) > 0$. Completing the square,

$$\mathrm{Var}(Y - \theta X) = \mathrm{Var}(Y) - \frac{c^2}{v_X} + v_X \left(\theta - \frac{c}{v_X}\right)^2.$$

The minimum is at $\theta^* = c/v_X$ with value $\mathrm{Var}(Y)(1 - \rho_{YX}^2)$. Vector version: $\beta^* = \Sigma_X^{-1} c$, residual R with $E[XR] = 0$ and $\mathrm{Var}(R) = \mathrm{Var}(Y) - c'\Sigma_X^{-1} c$.

Remember. Projection is the unique optimal linear adjustment.

The adjusted estimator stays centred

The Master 2 toolkit MASTER 2

From intuition: the lottery balances last year's scores on average too.

With $Y_i^* = Y_i - \beta'(X_i - \bar{X})$ for a *fixed* β,

$$\hat{\tau}_{adj} = \bar{Y}_1 - \bar{Y}_0 - \beta'(\bar{X}_1 - \bar{X}_0).$$

Since X is pre-treatment, randomisation gives $E_D[\bar{X}_1 - \bar{X}_0] = 0$, hence $E_D[\hat{\tau}_{adj}] = \tau$. When β is estimated on the same data, exact unbiasedness is lost; the fully interacted OLS estimator is consistent and asymptotically no less efficient than $\hat{\tau}$ (Lin, 2013).

Remember. Estimated adjustment is justified asymptotically or by sample splitting.

Cluster design effect

The Master 2 toolkit MASTER 2

From intuition: a shock shared by the whole class does not average out.

Let $Y_{ij} = \mu + a_j + \varepsilon_{ij}$, clusters independent, components mutually independent with variances $\sigma_a^2, \sigma_\varepsilon^2$. For cluster size m,

$$\mathrm{Var}(\bar{Y}_j) = \sigma_a^2 + \frac{\sigma_\varepsilon^2}{m}.$$

Relative to m independent units with the same marginal variance,

$$\frac{\sigma_a^2 + \sigma_\varepsilon^2/m}{(\sigma_a^2 + \sigma_\varepsilon^2)/m} = 1 + (m-1)\rho, \qquad \rho = \frac{\sigma_a^2}{\sigma_a^2 + \sigma_\varepsilon^2}.$$

Remember. Repeated observations cannot average away a shock common to the cluster.

ITT under noncompliance

The Master 2 toolkit MASTER 2

From intuition: "keep all 100 offered students together".

Let Z be the random offer, $D(z)$ attendance and $Y(z,d)$ the outcome. Randomisation identifies $\mathrm{ITT}_Y = E[Y(1,D(1)) - Y(0,D(0))]$ always. Under exclusion $Y(z,d) = Y(d)$,

$$\mathrm{ITT}_Y = E\big[(Y(1) - Y(0))(D(1) - D(0))\big].$$

With monotonicity $D(1) \geq D(0)$, this equals $P(C)\,E[Y(1) - Y(0) \mid C]$ for compliers C: the basis of Session 5.

Remember. Noncompliance changes the estimand unless IV assumptions justify rescaling.

Attrition as post-assignment selection

The Master 2 toolkit MASTER 2

From intuition: "the control students we still observe are the stronger ones".

Let $R(z)$ indicate that the outcome is observed under assignment z. Randomisation gives

$$E[Y \mid Z = z, R = 1] = E\big[Y(z) \mid R(z) = 1\big].$$

The complete-case gap compares means over the different events $\{R(1) = 1\}$ and $\{R(0) = 1\}$, and generally differs from $E[Y(1) - Y(0)]$. Equal response rates do not imply equal responder populations. Lee (2009) bounds trim the arm with higher response under monotone response.

Remember. Missingness is a selection mechanism that the lottery did not randomise.

Bonferroni from the union bound

The Master 2 toolkit MASTER 2

From intuition: twenty outcomes, twenty chances to be lucky.

Let $\mathcal{H}_0$ be the true nulls and suppose each valid p-value satisfies $P(p_j \le u) \le u$. Rejecting when $p_j \le \alpha/m$,

$$P(\text{at least one false rejection}) \le \sum_{j \in \mathcal{H}_0} P\Big(p_j \le \frac{\alpha}{m}\Big) \le \frac{|\mathcal{H}_0|\,\alpha}{m} \le \alpha.$$

No independence between tests is required.

Remember. Familywise control follows from a bound on the whole family of tests.

Exact randomisation p-values

The Master 2 toolkit MASTER 2

From intuition: the 6-out-of-20 calculation with Awa, Chloé and Karim.

Let $\mathcal{A}$ be the allowed assignments with probabilities $\pi(a)$. Under the sharp null $Y_i(1) = Y_i(0)$ for all i, the observed Y is fixed across a, so

$$p_{rand} = \sum_{a \in \mathcal{A}} \pi(a)\, 1\{|T(a, Y)| \geq |T(a_{obs}, Y)|\}$$

is computable, and $P(p_{rand} \leq u) \leq u$ under the null. Discreteness can make the test conservative.

Remember. Exactness relies on the sharp null and the actual assignment mechanism.

Sharp and weak nulls differ

The Master 2 toolkit MASTER 2

From intuition: "no effect for anyone" is stronger than "no effect on average".

Sharp null: $\tau_i = 0$ for every i. Weak null: $N^{-1} \sum_i \tau_i = 0$.

Two students with effects $(+2, -2)$ satisfy the weak null but not the sharp null. Under the weak null alone, the observed outcomes cannot be reused unchanged for every hypothetical assignment, so the Fisher test is not exact for it; studentised permutation tests restore asymptotic validity (Wu and Ding, 2021).

Remember. An exact sharp-null test is not automatically an exact average-effect test.

Session 2 roadmap

1. Why a lottery?	Intuition
2. Replaying the lottery	Intuition
3. From the lottery to estimates	Notation
4. Planning: power and sample size	Notation
5. Sharper designs	Design
6. When the experiment is not clean	Design
7. The Master 2 toolkit	Master 2
8. Practice and wrap-up	**Synthesis**

Check of understanding

Practice and wrap-up SYNTHESIS

Two experiments each contain 1,000 students. One draws students; the other draws five schools.

1. Why can the second experiment be far less precise?
2. Why must students who declined tutoring stay in the offered group in the main analysis?
3. Which of the two designs better protects against note-sharing spillovers?

Remember. Explain the design before choosing a standard error.

Exercise: assess a tutoring trial

Practice and wrap-up

SYNTHESIS

Ten minutes, then compare explanations.

Treated mean 74, control mean 69, standard deviation 10 in each arm, 100 independent students per arm.

1. Compute the effect, standard error and approximate 95% interval.
2. With ideal CUPED and correlation 0.7, estimate the new standard error.
3. How many students per arm would detect 2.5 points with 80% power at 5%?
4. What changes if whole classes of 25 are drawn, with $\rho = 0.08$?

Remember. Distinguish the effect estimate from the uncertainty calculation.

Exercise: solution

Practice and wrap-up

SYNTHESIS

1. $\hat{\tau} = 5$; $SE = \sqrt{2} \approx 1.41$; interval $\approx [2.23, 7.77]$.
2. $SE \approx 1.41 \times \sqrt{0.51} \approx 1.01$.
3. $n_g \approx 2 \cdot 100 \cdot 2.8^2 / 2.5^2 \approx 251$ per arm.
4. Variance is multiplied by 2.92, so $SE \approx 1.41\sqrt{2.92} \approx 2.41$ and the interval roughly covers [0.3, 9.7]. Plan and infer at the class level.

Remember. The same 5-point estimate can be precise or vague depending on the design.

Derivation exercise: blocking and projection

Practice and wrap-up SYNTHESIS

From intuition: Part 5's two variance-reduction tools, computed exactly.

(a) Two independent strata have weights 0.6 and 0.4 and effect-estimator variances 4 and 9. Compute the variance of $\hat{\tau}_{str}$.

(b) Let $\mathrm{Var}(Y) = 100$, $\mathrm{Var}(X) = 25$, $\mathrm{Cov}(X, Y) = 30$. Derive θ^* and $\mathrm{Var}(Y - \theta^* X)$.

(c) Show that $\hat{V}$ is unbiased for $\mathrm{Var}_D(\hat{\tau})$ when the treatment effect is constant.

Remember. Use independence, the projection identity and Neyman's formula explicitly.

Derivation exercise: solution

Practice and wrap-up SYNTHESIS

(a) $0.6^2 \cdot 4 + 0.4^2 \cdot 9 = 1.44 + 1.44 = 2.88$.

(b) $\theta^* = 30/25 = 1.2$ and $\mathrm{Var}(Y - 1.2X) = 100 - 30^2/25 = 64$, i.e. $\rho^2 = 0.36$.

(c) Constant effect means $\tau_i = \tau$ for all i, so $S_\tau^2 = 0$ and $E_D[\hat{V}] - \mathrm{Var}_D(\hat{\tau}) = S_\tau^2/N = 0$.

Note that (a) is the variance of an *estimator*, (b) the variance of adjusted *outcomes*.

Remember. Do not confuse outcome variance with estimator variance.

What we can now say, step by step

Practice and wrap-up

SYNTHESIS

Intuition	A lottery cuts the arrow from motivation to treatment: right on average, noisy in any one draw.
Notation	$\hat{\tau} = \bar{Y}_1 - \bar{Y}_0$ is unbiased; $SE = \sqrt{s_1^2/n_1 + s_0^2/n_0}$; intervals; power and sample size.
Design	Blocking, CUPED and adjustment shrink noise; clusters inflate it; ITT, attrition, peeking and multiplicity need a plan.
Master 2	Neyman's exact variance, finite-population CLT, optimal allocation, projection, design effect, randomisation inference.

Remember. A randomised design is powerful when its intended comparison survives in practice.

Next session: when treatment was chosen, not drawn

Practice and wrap-up

SYNTHESIS

Most real decisions are not randomised. Can we recover a fair comparison anyway?

Session 1 showed that comparing like with like works under exchangeability given X. Session 3 turns this into practical estimators: matching, propensity scores, inverse weighting and doubly robust methods, with diagnostics for overlap and balance.

Guiding question: how close can observational data get to the lottery?

Remember. Without a lottery, the design must be argued rather than guaranteed.

Further reading

Practice and wrap-up SYNTHESIS

- Imbens, G. W. and Rubin, D. B. (2015). *Causal Inference for Statistics, Social, and Biomedical Sciences*. Cambridge UP. Ch. 4–7, 9.
- Gerber, A. S. and Green, D. P. (2012). *Field Experiments: Design, Analysis, and Interpretation*. Norton.
- Kohavi, R., Tang, D. and Xu, Y. (2020). *Trustworthy Online Controlled Experiments*. Cambridge UP.
- Deng, A., Xu, Y., Kohavi, R. and Walker, T. (2013). Improving the sensitivity of online controlled experiments by utilizing pre-experiment data. *WSDM*.
- Lin, W. (2013). Agnostic notes on regression adjustments to experimental data. *Annals of Applied Statistics* 7(1).
- Li, X. and Ding, P. (2017). General forms of finite population central limit theorems. *JASA* 112(520).

Remember. Imbens and Rubin, chapters 4–6, is the best companion to today's session.